

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 6

Case Study: Modelling and Forecasting Stock Return Volatility: IBOVESPA GARCH Analysis

Application area: *Finance — Equity Risk and Value-at-Risk*

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Yahoo Finance: $\hat{BVSP}$ (IBOVESPA daily closing prices, 2000–2024) via `quantmod` or `yfinance`. For multivariate extension: also download $\hat{GSPC}$ (S&P 500) to estimate DCC between Brazilian and US equities.

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1. Background and Objectives

The IBOVESPA is one of the most volatile major equity indices, reflecting Brazil's exposure to commodity cycles, political risk, and global risk-off episodes. This tutorial estimates GARCH-family models to characterise conditional volatility, then constructs VaR forecasts and backtests them against realised losses.

Learning objectives: (i) Test for ARCH effects; (ii) estimate GARCH(1,1), GJR-GARCH, and EGARCH; (iii) compare models using information criteria and residual diagnostics; (iv) construct daily VaR and backtest; (v) estimate DCC between IBOVESPA and S&P 500.

2. Data Preparation

```
library(rugarch); library(rmgarch); library(quantmod)
getSymbols("^BVSP", src="yahoo", from="2000-01-01", to="2024-12-31")
ret_ibov <- 100 * diff(log(Cl(BVSP)))[-1]
getSymbols("^GSPC", src="yahoo", from="2000-01-01", to="2024-12-31")
ret_sp <- 100 * diff(log(Cl(GSPC)))[-1]
```

3. Exercises

3.1. Stylised facts

Plot the IBOVESPA log-returns. Compute excess kurtosis. Apply the Engle ARCH-LM test at $q = 5$ and $q = 10$ lags. Plot the ACF of r_t^2 . What stylised facts are present?

3.2. GARCH(1,1) estimation

Estimate a GARCH(1,1) with Normal innovations using MLE (**rugarch**).

- Report $\hat{\omega}, \hat{\alpha}, \hat{\beta}$ and their standard errors.
- Compute the unconditional variance $\bar{\sigma}^2 = \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})$.
- Compute the half-life of volatility shocks: $\ln(0.5) / \ln(\hat{\alpha} + \hat{\beta})$ days.
- Plot the estimated conditional standard deviation $\hat{\sigma}_t$ against the return series.

3.3. Asymmetric GARCH

Estimate GJR-GARCH(1,1) and EGARCH(1,1) with t -distributed innovations.

- (a) Is the leverage parameter $\hat{\gamma}$ statistically significant in both models?
- (b) Compare AIC and BIC across GARCH(1,1), GJR, and EGARCH. Which is preferred?
- (c) Apply the Ljung–Box test to standardised residuals $\hat{z}_t$ and to $\hat{z}_t^2$. Do residuals pass?

3.4. Value-at-Risk

Using the preferred model estimated on 2000–2019 (in-sample):

- (a) Produce 1-day-ahead 99% VaR forecasts for 2020–2024 (out-of-sample).
- (b) Compute the violation rate (fraction of days when return $< -\text{VaR}$). Is it close to 1%?
- (c) Apply the Kupiec (1995) proportion-of-failures test. Do you reject correct coverage?
- (d) Plot the VaR series and mark violations. Are they clustered during crisis periods?

3.5. DCC: IBOVESPA and S&P 500

Estimate the DCC model (`rmgarch`) for IBOVESPA and S&P 500 returns jointly.

- (a) Plot the dynamic conditional correlation $\hat{\rho}_{12,t}$ over 2000–2024.
- (b) At what periods did correlations peak? Relate to global financial crises.
- (c) What are the implications for international portfolio diversification from a Brazilian investor's perspective?

4. Core References

Engle, R.F. (1982). *Econometrica*, 50, 987–1007. Bollerslev, T. (1986). *JoE*, 31, 307–327. Engle, R.F. (2002). *JBES*, 20, 339–350. Tsay, R.S. (2010). *Analysis of Financial Time Series*. Wiley.
